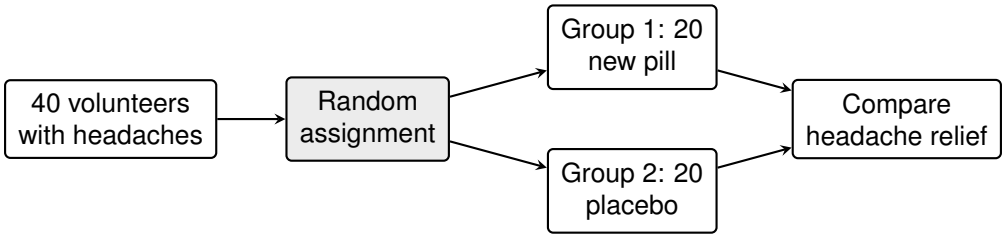
Experiments KEY

Name _____

Does a new pill stop headaches? We cannot just compare people who chose to take it with people who did not. They may be different in other ways. An **experiment** gives the treatment on purpose and compares fairly.

Word	**In the headache experiment**
Subjects (experimental units)	The 40 volunteers with headaches
Explanatory variable	What we change: the type of pill
Treatments	The things we compare: new pill, placebo
Response variable	What we measure: headache relief
Control group	The group that does not get the new treatment (the placebo group)
Placebo	A fake treatment that looks real, like a sugar pill
Double-blind	Neither the subjects nor the people measuring know who got which pill



This is a **completely randomized design**. Chance alone decides which group each subject joins.

Principles of a good experiment

Control	Compare to a control group. Keep everything else the same for both groups.
Random assignment	Use chance to put subjects into groups. Then the groups are about the same, except for the treatment.
Replication	Use many subjects. With more subjects, chance differences between the groups get smaller.

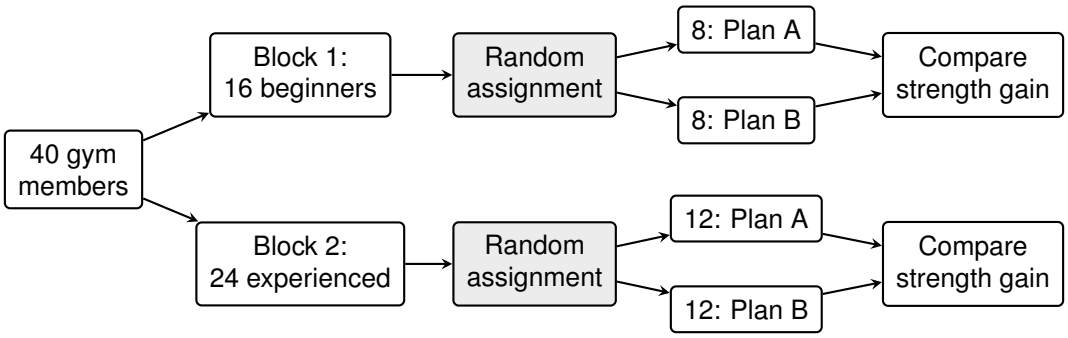
Why a placebo? People often feel better just because they *think* they got medicine (the **placebo effect**). With a placebo group, both groups think they are being treated.

Matched pairs design. Each subject gets **both** treatments, in a random order. Or pair up very similar subjects, like twins. Each twin gets a different treatment.

Example: Which running shoe is faster? Each runner runs once in each shoe. A coin flip decides which shoe goes first. Comparing each runner to themselves removes the differences between runners.

Randomized block design. First split the subjects into groups that are alike (**blocks**). Then randomly assign the treatments inside each block. Comparing inside each block keeps the differences between blocks out of the result.

Example: A gym tests two workout plans with 40 members. 16 are beginners and 24 are experienced. Beginners usually get stronger faster. The blocks are the beginners and the experienced members. Inside each block, chance decides who gets each plan.



Blocks and strata. A randomized block design works like a stratified sample (5.2). Both split people into groups that are alike, then use chance inside each group. In a stratified sample, chance picks **who is in the sample**. In a randomized block design, chance picks **which treatment** each subject gets.

Work with a partner. Compare answers when you finish each problem.

1. 60 tomato plants are randomly split into two groups of 30. One group gets a new fertilizer. The other gets only water. After 6 weeks, the weight of the tomatoes is measured.

   (a) What are the subjects? The 60 tomato plants

   (b) What is the explanatory variable? Fertilizer or water (the treatment)

   (c) What is the response variable? The weight of the tomatoes

   (d) What is the control group? The 30 plants that get only water

   (e) Why split the plants **randomly**? So the groups are alike. Then a difference comes from the fertilizer, not from which plants were stronger.

2. In the headache experiment, the study is double-blind.

   (a) Why should the subjects not know which pill they got?
   If they know they got the real pill, they might feel better just from believing it (the placebo effect).

   (b) Why should the nurses who measure headache relief not know?
   They might, without meaning to, rate the new-pill group better.

3. Does listening to music help students memorize a list of words? You have 50 student volunteers. Describe a completely randomized experiment. Name the treatments, how you assign them, and what you measure.

   Treatments: music, no music. Number the students 1 to 50 and use a random number generator to put 25 in each group. Both groups study the same list for the same time in the same room. Measure how many words each student remembers. Compare the two groups.

4. A company tests two hand lotions. Each of 20 volunteers puts Lotion A on one hand and Lotion B on the other. A coin flip decides which hand gets A. After a week, a doctor rates the dryness of each hand.

   (a) What kind of design is this? Matched pairs

   (b) Why flip a coin to pick the hand? Most people use one hand more. The coin keeps that from favoring one lotion.

   (c) Why is this design better than giving 10 people Lotion A and 10 people Lotion B?
   Each person is compared to themselves, so differences in skin from person to person do not matter.

5. A teacher tries a new reading program with her 1st-hour class and the old program with her 6th-hour class. The 1st-hour class scores higher. Can she conclude the new program works better? Explain.

   No. She did not use random assignment. The classes could differ in other ways (time of day, which students take 1st hour). Those differences could cause the higher scores.

6. A school tests a reading program. It splits students by grade, then randomly assigns half of each grade to the program.

   (a) What kind of design is this? A randomized block design

   (b) Why split by grade first?
   Grade could affect reading scores. Comparing within each grade keeps that difference out of the result.